

Pretrained-Embedding Node Initialization for Heterogeneous-Graph Semi-Supervised Short-Text News Classification

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Abstract—Short-text news classification is difficult because headlines are semantically sparse and labeled data are scarce. Heterogeneous graph neural networks address sparsity by connecting documents to auxiliary nodes (topics and entities) so that unlabeled documents can share information through a corpus-level graph. This paper does not propose a new architecture; instead it studies a single, isolable design choice in the Heterogeneous Graph Attention Network (HGAT) of Linmei et al.: how the graph’s nodes are initialized. We keep the HGAT graph schema and the dual-level (node- and type-level) attention fixed, and replace the standard high-dimensional TF-IDF/one-hot node features with low-dimensional pretrained word2vec embeddings for entity nodes and pooled word-embedding vectors for document nodes, retaining Latent Dirichlet Allocation (LDA) topic distributions for topic nodes. On a 6,000-document AG News subset and a 6,000-document HuffPost News Category subset, the embedding-initialized model reaches 76.30% and 59.57% accuracy in a transductive semi-supervised setting with few labels per class, improving over the TF-IDF-initialized HGAT by 4.20 and 2.82 points respectively. We report these results transparently as single-run measurements and discuss their limitations candidly. This manuscript documents work completed in 2021 as part of the author’s M.Sc. thesis; its intent is a faithful, honest record of that work: a corrected and reproducible specification of the pipeline, and a clear-eyed analysis, grounded in the released code, of what the method actually computes.

Index Terms—Short-text classification, heterogeneous graph neural network, graph attention network, semi-supervised learning, node feature initialization, word embeddings, topic modeling, entity linking.

I. Introduction

NEWs classification—assigning a short article or headline to a topical category such as World, Sports, or Business—is a canonical short-text classification task. Two properties make it hard. First, headlines are short: a handful of words offer little co-occurrence signal, so bag-of-words representations are sparse and the same concept surfaces under many surface forms. Second, labels are scarce: new categories and events appear continuously, and manually labeling a representative sample is costly. These properties motivate semi-supervised methods that exploit a large pool of unlabeled documents alongside a small labeled set.

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Graph neural networks (GNNs) are a natural fit. By representing a corpus as a graph and propagating information along edges, a GNN lets labeled and unlabeled documents share evidence. Text GCN [1] models a corpus as a document–word graph; the Heterogeneous Graph Attention Network (HGAT) [2], [3] enriches this idea by adding topic nodes (from Latent Dirichlet Allocation, LDA [4]) and entity nodes (from Wikipedia entity linking [5]), yielding a heterogeneous information network on which a dual-level attention mechanism aggregates neighbors by both node importance and node type.

A recurring, under-examined question in this pipeline is how the graph’s nodes are initialized. HGAT and its predecessors initialize document and entity nodes with high-dimensional sparse features (TF-IDF or one-hot). Dense pretrained embeddings such as word2vec [6] are lower-dimensional and encode distributional semantics, and are therefore a plausible drop-in improvement, particularly under label scarcity where the first-layer parameter count matters. This paper isolates exactly that choice.

Scope and contributions: We are deliberately conservative about what we claim. We do not introduce a new model, and—correcting an overstatement in the underlying thesis—we make no claim of a new attention mechanism: the attention used here is exactly the node- and type-level attention of HGAT [2], which we credit in full and specify precisely in Section III. Our contributions are:

- 1) An isolated study of node-feature initialization. Holding the HGAT graph and architecture fixed, we replace TF-IDF node features with pretrained-embedding features for entity and document nodes, and report the effect on two short-text news corpora (Section IV).
- 2) A corrected, reproducible pipeline specification. We give complete graph-construction and model equations, describe the transductive training and model-selection protocol exactly, and use consistent notation throughout (Sections III–IV).
- 3) An honest empirical account. We report single-run results and discuss their limitations openly (Section VI), so that the scope of the evidence is clear.

The intent is deliberately modest. The contribution is a lightweight, training-free change to a single stage of an existing pipeline, and the scientific question it addresses

is a clean one: does dense pretrained initialization help a fixed heterogeneous text graph under label scarcity? We report the answer we observed on our data and are careful to state what the single-run setting can and cannot establish.

II. Related Work

We organize the prior work that this study builds on into three threads. As this manuscript documents work completed in 2021, the discussion is scoped to methods available at that time.

A. Graph neural networks for text classification

Spectral GCNs [7] and their spatial successors brought semi-supervised node classification to text. Text GCN [1] built a single document–word graph over the corpus; GraphSAGE [8] introduced inductive neighborhood sampling; and GAT [9] added masked self-attention so neighbors contribute unequally. For heterogeneous graphs, HAN [10] introduced meta-path-based node- and semantic-level attention, while HGAT [2], [3]—the model we build on—applied a meta-path-free dual-level attention to a document/topic/entity graph specifically for short text. Related graph formulations include TextING [11], which builds a per-document graph for inductive classification, and HyperGAT [12], which uses hypergraph attention to capture high-order word interactions. Among these, HGAT is the natural base for short news text because its topic and entity nodes directly target the semantic sparsity of headlines.

B. Semi-supervised and low-label text classification

Because labels are scarce, semi-supervised methods that exploit unlabeled data are central. Latent-variable and embedding-based approaches such as PTE [13] model documents, words, and labels jointly and learn node embeddings for classification. Consistency-regularization methods—UDA [14] and MixText [15]—augment a small labeled set with a large unlabeled pool. Graph-based transductive learning, as in HGAT, belongs to this family: the whole corpus graph is available at training time, so unlabeled and test documents provide structural context even though their labels are withheld.

C. Text representations and node features

The features used to initialize graph nodes matter. Classical bag-of-words and TF–IDF representations are high-dimensional and sparse. Distributed word representations—word2vec [6]—are dense, low-dimensional, and encode distributional semantics; a document can be summarized by pooling its word vectors, for example a TF–IDF-weighted centroid [16]. Contextual encoders such as BERT [17] and Sentence-BERT [18] were emerging at the time and are natural stronger instantiations of the same idea; we note them as directions rather than baselines here. This paper studies the simplest

such change—replacing sparse TF–IDF node features with static word2vec-based features—while keeping everything else in HGAT fixed.

III. Method

A. Overview and problem setting

Let $\mathcal{D} = \{d_1, \dots, d_n\}$ be a corpus of short documents, of which a small labeled subset $\mathcal{D}_L$ has labels in $\{1, \dots, C\}$. We build a single heterogeneous graph $G = (V, E)$ over the whole corpus and cast classification as transductive node classification of the document nodes. Fig. 1 shows the pipeline: entity and topic extraction, graph construction, node-feature initialization, and the HGAT classifier.

B. Heterogeneous graph construction

The graph contains three node types—document (D), topic (T), and entity (E)—and three undirected edge types, $D-T$, $D-E$, and $E-E$ (Fig. 2). Formally, with a node-type map $\phi : V \rightarrow \{D, T, E\}$ and an edge-type map, the network is heterogeneous because $|\{D, T, E\}| > 1$.

Entity nodes are produced by TAGME [5], which links substrings of a document to Wikipedia pages and returns, per annotation, a linking confidence ρ and a link probability. Documents are connected to their entities ($D-E$), and two entities are connected ($E-E$) when their pretrained-embedding cosine similarity exceeds a threshold, so that documents sharing related entities become close in the graph.

Topic nodes are produced by LDA [4] trained with gensim [19]. Each document is connected to its top- P topics by posterior probability ($D-T$). The number of topics K is chosen by topic coherence [20].

These edges induce the meta-paths $D-T-D$, $D-E-D$, and $D-E-E-D$, along which two documents are considered semantically related.

C. Node-feature initialization (the studied component)

Each node type receives an initial feature vector x_v . This is the only stage we change relative to [2].

- Entity nodes. The entity’s surface text is mapped to its 300-dimensional Google-News word2vec [6] vector. Entities whose surface form is out-of-vocabulary are dropped together with their edges.
- Document nodes. The document is tokenized and each in-vocabulary token is mapped to its word2vec vector; the document feature is a centroid of these vectors. The intended pooling is a TF–IDF-weighted centroid [16]; we note in Section VI that the released code uses the unweighted mean.
- Topic nodes. The LDA topic–word distribution is used directly as the topic feature.

Because node types have different feature dimensionalities, the model uses type-specific input projections (below).

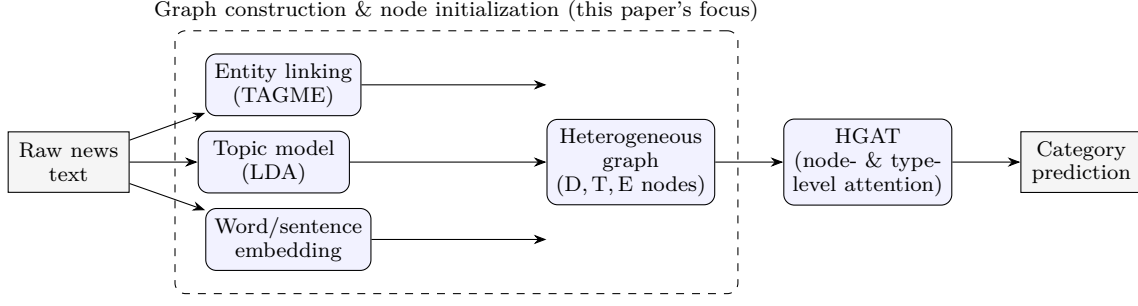


Fig. 1. Overall pipeline. The graph schema and the HGAT classifier are inherited unchanged from [2]; this paper isolates the node-initialization stage (entity/document features), highlighted by the dashed box.

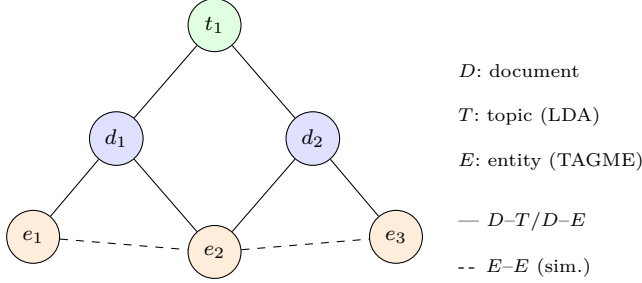


Fig. 2. Heterogeneous graph schema. Documents connect to topics and entities; similar entities connect to each other, linking documents via $D-E-E-D$ meta-paths.

D. HGAT: dual-level attention

We restate the HGAT propagation of [2] in full and in consistent notation; the mechanism is theirs, not ours. Let $\tau \in \{D, T, E\}$ index node types, let H_τ collect the features of type- τ nodes, and let W_τ be a type-specific projection. Let $\tilde{A}$ be the symmetrically normalized adjacency with self-loops, $\tilde{A} = \hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$.

Type-level attention. For a target node v of type τ' , the contribution of neighbor type τ is summarized by

$$h_v^\tau = \sum_{v' \in \mathcal{N}_v \cap \tau} \tilde{A}_{vv'} x_{v'} W_\tau, \quad (1)$$

and a type-attention weight is computed and normalized over types:

$$a_v^\tau = \sigma(\mu^\top [x_v W_{\tau'} \parallel h_v^\tau]), \quad \alpha_v^\tau = \frac{\exp(a_v^\tau)}{\sum_{\tau''} \exp(a_v^{\tau''})}, \quad (2)$$

where μ is a learned attention vector, $\parallel$ is concatenation, and $\sigma(\cdot)$ is a nonlinearity.

Node-level attention. Within each type, neighbors are weighted by an additive (GAT-style) attention. For projected features $h = x_v W_{\tau'}$ and $g = x_{v'} W_{\tau'}$,

$$e_{vv'} = \text{LeakyReLU}(a_1^\top h + a_2^\top g), \quad \beta_{vv'} = \text{softmax}_{v'}(e_{vv'}), \quad (3)$$

with learned vectors a_1, a_2 . Following the reference implementation, the effective propagation weight blends attention with the normalized adjacency,

$$\hat{\beta}_{vv'} = \gamma \beta_{vv'} \deg(v) + (1 - \gamma) \tilde{A}_{vv'}, \quad (4)$$

where $\gamma \in [0, 1]$ controls how much learned attention departs from plain GCN propagation (γ is small in the reference code, so the adjacency dominates message passing; see Section VI).

Layer update and classifier. A layer aggregates the type-attended messages,

$$h_v^{(l+1)} = \rho \left(\sum_{\tau} \alpha_v^\tau \sum_{v' \in \mathcal{N}_v \cap \tau} \hat{\beta}_{vv'} x_{v'}^{(l)} W_\tau^{(l)} \right), \quad (5)$$

with $\rho = \text{ReLU}$ and dropout between the two layers. The second layer maps to C logits per document node, and training minimizes the cross-entropy over labeled document nodes,

$$\mathcal{L} = - \sum_{v \in \mathcal{D}_L} \sum_{c=1}^C y_{vc} \log \text{softmax}(z_v)_c. \quad (6)$$

E. Transductive training

The full graph—including unlabeled and test document nodes—is used during training, but test labels are never used; test nodes act as unlabeled context. Optimization uses Adam [21]. We report test accuracy at the epoch of highest validation accuracy over a fixed 300-epoch budget; the test set is used only for final reporting. We discuss the transductive nature (and its consequences for corpus-level statistics) in Section VI.

IV. Experimental Setup

A. Datasets

We use two balanced short-text news corpora. AG News [22]: a random 6,000-document subset with four evenly sized classes (World, Sports, Business, Sci/Tech). HuffPost News Category (5-class subset) [23]: a random subset of the Kaggle News Category Dataset (HuffPost headlines) restricted to five evenly sized classes (Politics, Food, Sports, Crime, Science), 1,200 documents each.

We stress a naming correction: the underlying thesis referred to the second corpus as “TagMyNews.” It is not the TagMyNews benchmark of Vitale et al. [24] used by [2]; it is the HuffPost News Category Dataset [23]. We rename it accordingly and do not compare its numbers against published TagMyNews results. Table I gives the constructed-graph statistics.

TABLE I
Constructed-graph statistics (single construction).

Dataset	Doc	Entity	Topic	Train	Val	Test
AG News	6000	10298	15	160	160	5680
HuffPost-5	6000	6898	17	800	800	4400

TABLE II
Accuracy (%) with pretrained-embedding vs. TF-IDF node features. Single run; see Section VI for the caveats on each row.

Dataset	Node features	Acc	P	R	F1
AG News	TF-IDF [2] [†]	72.10	–	–	–
	Embeddings (ours)	76.30	76.4	76.3	76.3
HuffPost-5	TF-IDF [‡]	56.75	56.8	56.9	56.7
	Embeddings (ours)	59.57	59.9	59.6	59.5

[†]Reported in [2] on a different subset with 20 train + 20 val per class.

[‡]Base HGAT re-run on this corpus at the same budget. Absolute gains: +4.20 and +2.82 points.

B. Preprocessing and graph hyperparameters

Text is lowercased, tokenized, stop-word- and symbol-filtered, and lemmatized; multiword entities are joined by underscores. Entity nodes are pruned by TAGME confidence; entity–entity edges require word2vec cosine similarity > 0.5 ; each document is linked to its top $P=2$ topics. Two graph-construction details (the exact ρ threshold applied at construction versus pruning, and an entity minimum-degree rule) are discussed in Section VI.

C. Model hyperparameters and protocol

Two-layer HGAT, hidden dimension 512, Adam with learning rate 0.005, weight decay 5×10^{-8} , dropout 0.95 (a genuine drop rate, inherited unchanged from the reference implementation; Section VI), 300 epochs, test-at-best-validation model selection. Accuracy and macro precision/recall/F1 are computed on document nodes. All numbers below are from a single run with a fixed seed.

V. Results and Analysis

A. Main comparison

Table II reports the embedding-initialized model against the TF-IDF-initialized HGAT. On AG News, accuracy is 76.30% versus 72.10%; on HuffPost-5, 59.57% versus 56.75%. The two rows differ in how the baseline was obtained, and we state this openly. The AG News baseline figure is taken from [2], which used a different random subset and a smaller labeled budget (20 vs. 40 train per class), so the AG News comparison is not fully controlled. The HuffPost-5 row is a controlled comparison: the base method was re-run on the identical corpus at the identical 160-labels-per-class budget (both rows carry full precision/recall/F1), so the only uncontrolled factor there is that, like all numbers in this study, it is a single run. We return to both points in Section VI.

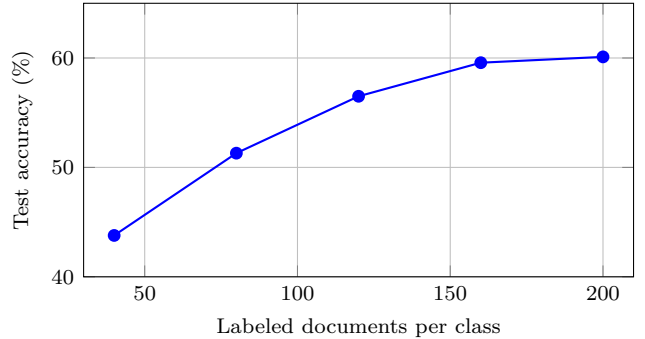


Fig. 3. Label-efficiency curve on HuffPost-5 (single run). Accuracy saturates near 160 labels/class; the plateau requires multi-seed confirmation.

B. Label efficiency

Fig. 3 plots HuffPost-5 accuracy against the number of labeled documents per class (single run). Accuracy rises steeply from 43.78% at 40/class to 59.57% at 160/class and then flattens (60.10% at 200/class), consistent with the intuition that the graph lets a modest number of labels propagate widely. The final 160 $\rightarrow$ 200 difference (+0.53) is small enough to be within run-to-run noise, so we read the curve qualitatively rather than treating the plateau as established.

C. Interpreting the model

Two points help interpret the numbers. First, the node-level attention of Eq. (4) blends learned attention with the normalized adjacency; in the reference configuration the small γ means the adjacency dominates, so attention modulates only a fraction of message passing. Second, the document feature is a centroid of word vectors, so word order is discarded and the representation is close to a continuous bag-of-words; the heterogeneous graph is what is expected to add discriminative structure beyond this pooled feature. Both observations are noted here so the reported gains are not over-interpreted, and both point to natural follow-up experiments (Section VI).

VI. Limitations and Future Work

We state the limitations of this study openly, in the spirit of an honest record rather than an over-claim.

- 1) Single-run results. Training uses a single fixed seed (42) and a single train/test split, so no variance estimate is available. Transductive GNNs at small label budgets are known to vary by a few points across seeds and label draws [25]; the reported differences should therefore be read as indicative. Repeating each configuration over multiple seeds and label-split resamples, and reporting mean $\pm$ std with a significance test, is the most valuable strengthening of this work.
- 2) AG News baseline provenance. The AG News 72.10% figure is quoted from [2] on a different subset and a smaller labeled budget, so the AG News comparison

is not fully controlled. The HuffPost-5 comparison, by contrast, is a matched re-run of the base method on the same corpus and budget. Re-running the TF-IDF baseline on the identical AG News graph and split would make that comparison controlled as well.

- 3) No isolating ablation. A factorial ablation—{TF-IDF, mean word2vec, TF-IDF-weighted centroid} $\times$ {entity-feature change, document-feature change}—together with a no-graph linear classifier on the document features, would separate the contribution of the feature change from that of the graph. This is the natural next experiment.
- 4) Document pooling. The released code computes an unweighted mean of word vectors for document nodes rather than the TF-IDF-weighted centroid [16] described; reporting both is straightforward and would clarify which pooling drives the result.
- 5) Transductive setting. LDA topics, corpus statistics, and entity edges are computed over the whole corpus, which includes the test documents (their labels are never used). This is standard for transductive graph learning but means the numbers describe a fixed corpus; an inductive variant (statistics and graph built without test documents, attached at inference) would be needed for streaming deployment.
- 6) Regularization. Dropout is a genuine drop rate of 0.95 and weight decay is 5×10^{-8} , both inherited unchanged from the reference implementation. These are aggressive and were not tuned here; a small sensitivity study would confirm whether they are appropriate for this data.
- 7) Graph construction. The ρ confidence threshold is applied at two pipeline stages (a permissive filter at construction, a stricter one at pruning); an entity minimum-degree rule ensures connectivity. Reporting entity counts at each stage and a sensitivity analysis over these thresholds and the topic count K would make the graph fully reproducible from the text.
- 8) Unmeasured efficiency. Low-dimensional dense features reduce the first-layer parameter count relative to vocabulary-sized TF-IDF, but this study does not measure training time, parameter counts, or memory; we therefore make no efficiency claim.
- 9) Stronger encoders. Contextual encoders (BERT, Sentence-BERT [17], [18]) are natural stronger instantiations of the document-embedding idea and were emerging at the time; comparing them under the same fixed graph is a clear direction for follow-up work.

VII. Conclusion

This paper documents a 2021 study of a heterogeneous-graph short-text news classifier, isolating one design choice—pretrained-embedding node initialization—while keeping the HGAT architecture and its dual-level attention fixed and fully credited to Linmei et al. On two 6,000-

document news corpora, replacing sparse TF-IDF node features with low-dimensional word2vec-based features improved single-run accuracy over the TF-IDF-initialized baseline by 4.20 points on AG News and 2.82 points on HuffPost-5, the latter under a matched budget. We have reported these results transparently and discussed their limitations—chiefly the single-run setting and the AG News baseline provenance—without overstating them. The lasting value of the work is a clean, reproducible specification of a document/topic/entity graph pipeline and a candid account of how a simple, training-free change to node initialization behaves under label scarcity. As a record of the author’s M.Sc. thesis, it is offered as an honest snapshot of that work; the natural continuation—multi-seed evaluation, an isolating ablation, and stronger encoders under the same graph—is left to subsequent study.

Data Availability

The two corpora derive from public sources: AG News [22] and the Kaggle News Category Dataset (HuffPost) [23]. The training code and the AG News data used are available in the project repository.

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